

CJ 540

EXPERIMENTS

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EXPERIMENTS INVOLVE

OBSERVING

measuring an outcome of interest

TAKING ACTION

having participants do something, or doing something to participants

OBSERVING AGAIN

measuring changes in the outcome of interest
(consequences of the action)

EXPERIMENTS AND CAUSATION

EXPERIMENTS DESIGNED FOR EXPLANATORY (CAUSAL) PURPOSES

- Testing hypotheses



Example: You hypothesize that learning about Muslim contributions to U.S. history will reduce prejudice.

To test hypothesis in an experiment, you would:

- measure prejudice
- expose participants to documentary/videos that depict Muslim contributions
- measure prejudice, again (to see if it reduced)

THE CLASSICAL EXPERIMENT



Independent and
Dependent Variables



Pre-testing and
Post-testing



Experimental and Control
Groups

INDEPENDENT AND DEPENDENT VARIABLES

INDEPENDENT

A **stimulus** or cause

- dichotomous (either present or absent, exposure or not)

DEPENDENT

An **outcome** or effect

PRE-TESTING

Measuring the dependent variable (outcome) among participants before they are exposed to the stimulus

POST-TESTING

Measuring the dependent variable (outcome) among participants after they are exposed to the (present or absent) stimulus

PRE-TESTING AND POST-TESTING

EXPERIMENTAL AND CONTROL GROUPS

EXPERIMENTAL GROUP

Participants **who are exposed** to the stimulus

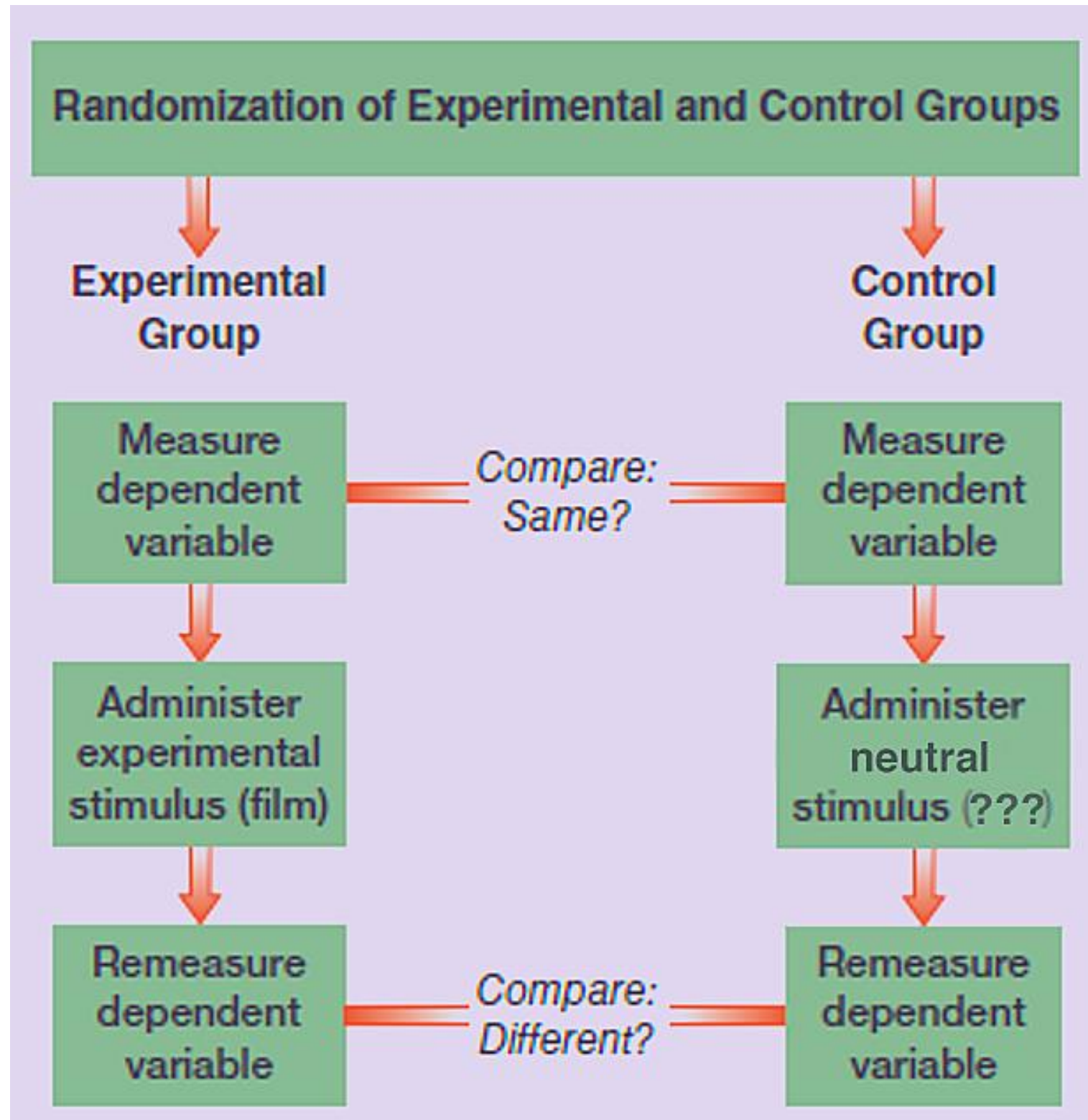
- the independent variable is present

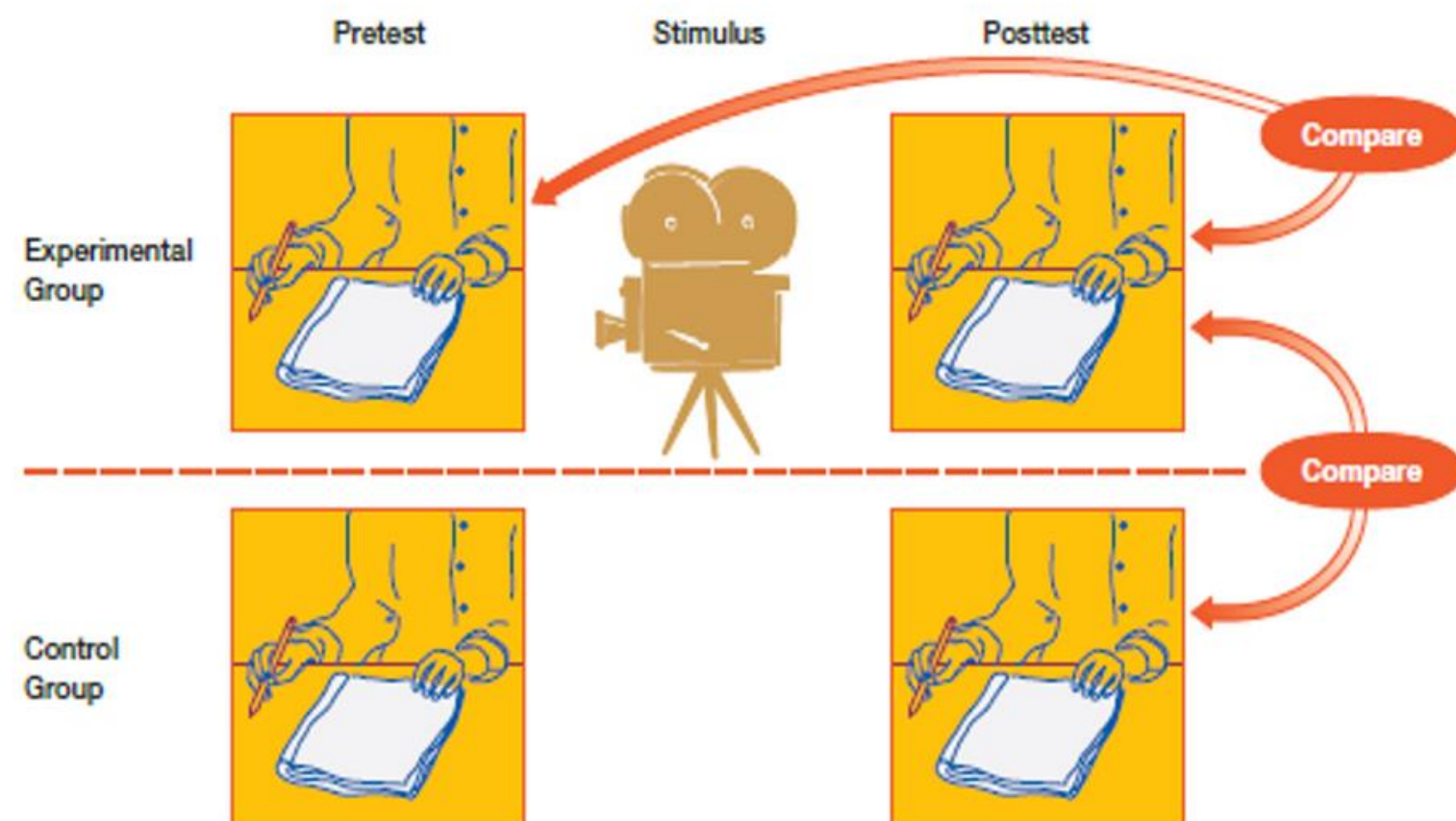
CONTROL GROUP

Participants who resemble the experimental group in all respects, but **who are not exposed** to the stimulus

- Often exposed to placebo/something neutral that is not intended to affect outcome

THE CLASSICAL EXPERIMENT





THE CLASSICAL EXPERIMENT

AN EXAMPLE

A version of the classical experiment in which neither the participants nor the researcher know who is in which group



THE DOUBLE-BLIND EXPERIMENT

EXPERIMENTS, THE UNIVERSITY, AND GENERALIZABILITY

We want social research findings that are generalizable to the population. This means that the sample must be representative of the population.

However, because many researchers are university professors, their experiments tend to rely on undergraduate students as participants

How would we ensure that our findings are generalizable?

If not possible, how do we ensure our findings are valid?

Selecting Participants

(for representativeness)

PROBABILITY SAMPLING

Representative if:

- Sample size is large enough
- Entire sample of participants comes from probability sampling frame
- Each group (experimental and control) is drawn from participants based on probability sampling frame

Selecting Participants

(for similarity between groups)

RANDOM ASSIGNMENT

Randomly assigning participants to either experimental group or control group

- Use random number generator to assign people to each group based on number
- Participants have equal chance of being in either group

Selecting Participants

(for similarity between groups)

QUOTA MATCHING*

Selecting participants' group in
attempt to match relevant

characteristics across groups

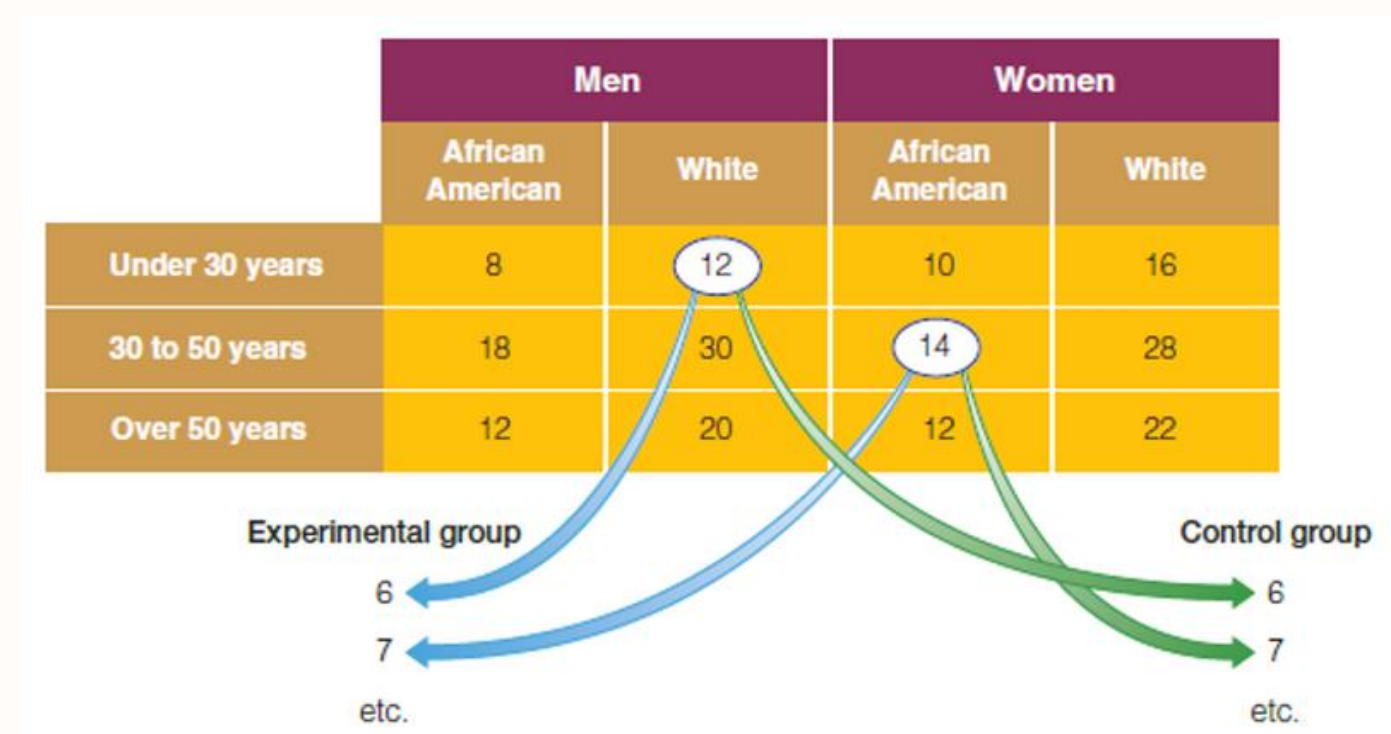
- Create quota matrix of all relevant characteristics (e.g. sex, age, race, etc.)
- Divide total in each cell of quota matrix equally between experimental and control groups

*least preferred method of participant selection

Selecting Participants

(for similarity between groups)

QUOTA MATCHING*



*least preferred method of participant selection

BEST METHODS FOR SELECTING PARTICIPANTS

BEST

Probability Sampling + Random
Assignment

SECOND BEST

Random Assignment (within Non-
Probability Sample)

THIRD BEST

Matching (within Non-Probability Sample)

VALIDITY ISSUES IN EXPERIMENTS

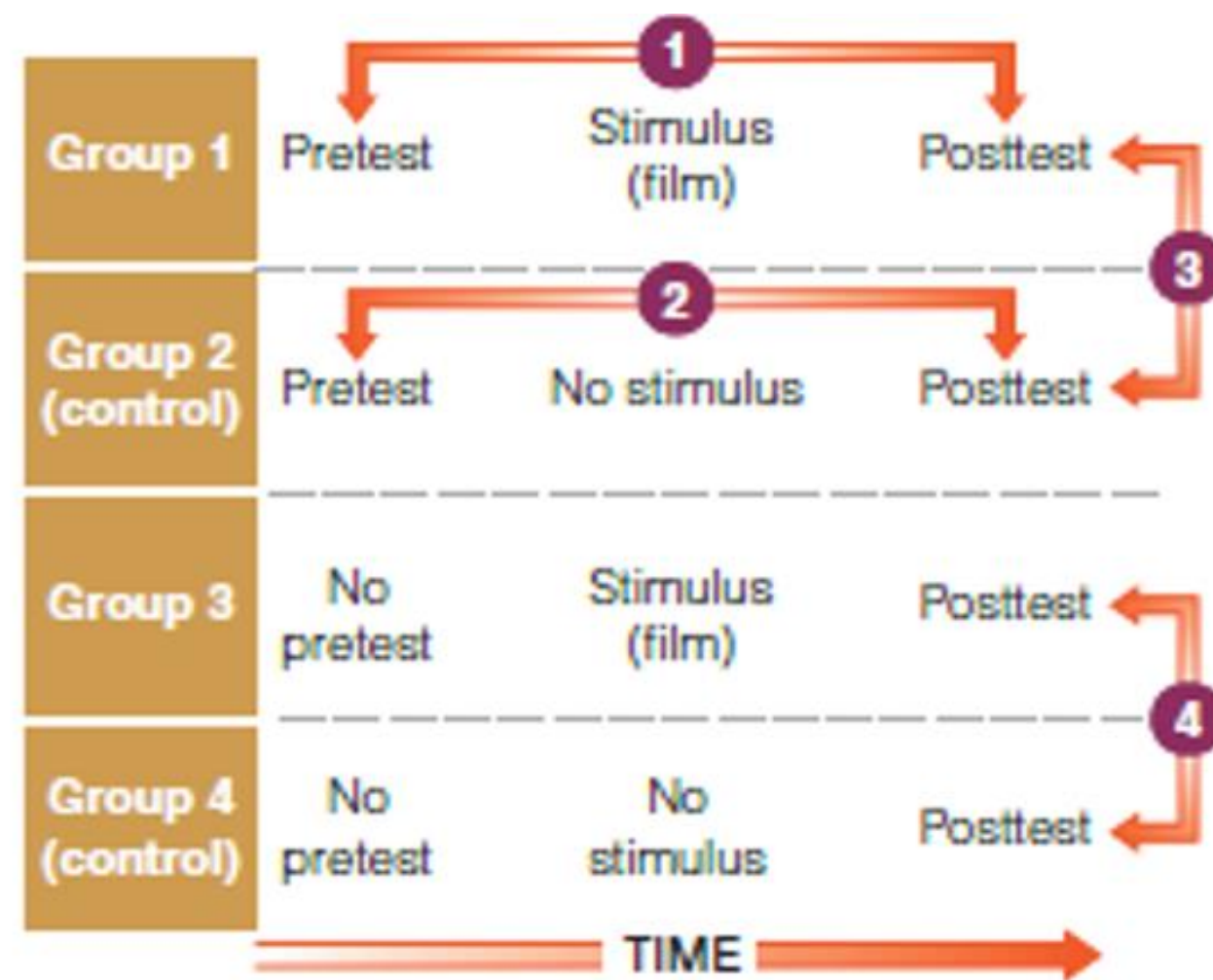
INTERNAL VALIDITY ISSUES

Possibility that anything other than the stimulus can affect the outcome

- history, maturation, testing, instrumentation, statistical regression, selection bias, experimental mortality, causal time order, diffusion or imitation of treatments, compensation, compensatory rivalry, demoralization

EXTERNAL VALIDITY ISSUES

Possibility that results may not be generalizable to the "real" world



Expected Findings

- 1 In Group 1, posttest prejudice should be less than pretest prejudice.
- 2 In Group 2, prejudice should be the same in the pretest and the posttest.
- 3 The Group 1 posttest should show less prejudice than the Group 2 posttest does.
- 4 The Group 3 posttest should show less prejudice than the Group 4 posttest does.

ADDRESSING VALIDITY ISSUES

THE SOLOMON FOUR-GROUP DESIGN

TYPES OF CLASSICAL EXPERIMENTS

LAB EXPERIMENT

Experiment in which participants are required to come into the lab to participate in the research study.

Classical Experiment traditionally seen as a Lab Experiment

FIELD EXPERIMENT

Experiment that take place outside the lab, in the real world.

Stimulus and measurement instruments (often surveys) are sent to participants via **mail** or **web** (internet links, websites, emails)

Field experiments, especially web-based experiments are currently most common

NATURAL EXPERIMENT

Experiment in which stimulus (sometimes called the treatment) has already naturally-occurred.

Quasi-Experimental because lacks pre-test comparison, and stimulus not under researcher control.

Used in disaster research, effect of disaster on those who experienced it

A QUASI-EXPERIMENT

STRENGTHS

- Causation: Isolation of independent variable's/stimulus' impact on outcome
- Replication

WEAKNESSES

- Artificiality: External validity problems

STRENGTHS AND WEAKNESSES OF EXPERIMENTAL DESIGNS

Ethics and Experiments

- Deception for the stimulus, requires debriefing
- Obtrusive: has potential to inadvertently harm participants